

Chapter 1

Storytelling: The Historical Evolution of Disasters in Central Vietnam

While Chapter 5 presented the technical implementation of PCA, this chapter interprets those findings through a historical and socio-economic lens. By integrating the structural insights from the Biplot with a temporal trend analysis, we uncover a fundamental shift in the nature of natural disasters in Central Vietnam: a transition from humanitarian tragedies to economic burdens.

1.1 Economic Context: The Inflation Adjustment

To accurately compare disaster impacts across different eras, it is crucial to account for the time value of money. The dataset utilized in this study employs **Inflation-Adjusted Costs** (Real Value), calculated based on the Consumer Price Index (CPI). The adjustment formula is defined as follows [2]:

$$\text{Damage}_{\text{Adj}} = \text{Damage}_{\text{Nominal}} \times \frac{\text{CPI}_{\text{Ref}}}{\text{CPI}_{\text{Event}}} \quad (1.1)$$

Table 1.1 illustrates the magnitude of this adjustment for the key historical outliers identified in our analysis. This conversion reveals that historical storms, while seemingly less costly in nominal terms, inflicted massive economic destruction relative to the economy of that era.

Table 1.1: Major Historical Disasters in Central Vietnam (Verified Data)

Event / Year	Impact Zone	Era Context	Total Deaths	Damage (Adj. 2023)
Typhoon Iris (1964)	Central Coast	War / Low Infrastructure	7,000	\$471,770,000
Historic Flood (1999)	Thua Thien Hue	Pre-modern Urbanization	≈ 595	$\approx \$500,000,000$
Typhoon Xangsane (2006)	Da Nang City	Rapid Urban Growth	72	$\approx \$1,200,000,000$
Typhoon Molave (2020)	Quang Ngai	Modern Coastal Economy	≈ 23	$\approx \$850,000,000$

Source: Extracted from dataset analysis and Table 3.2 of this report [1].

1.2 Deep Analysis of PCA Visualizations

Our storytelling is constructed through a three-step analytical process, connecting the reliability of the model (Scree Plot) to the structure of destruction (Biplot) and finally to the historical evolution (Time Series).

1.2.1 Graph 1: Scree Plot – Establishing Reliability

The Scree Plot (Figure 5.1) is not merely a technical output; it establishes the **credibility** of our narrative.

- **Result:** The first two principal components (PC1 and PC2) cumulatively explain **70.4%** of the dataset’s variance.
- **Analysis:** This high retention rate implies that the complex history of disasters (involving four distinct variables: Deaths, Injured, Affected, Damage) can be reliably compressed into a 2-dimensional story. It confirms that the trends discussed below are dominant structural signals, not random statistical noise.

1.2.2 Graph 2: The Biplot – The “Map of Destruction”

The Biplot (Figure 5.2) serves as the compass for our analysis. It defines *what* makes a disaster severe and *how* different disasters behave.

Axis Interpretation

- **PC1 (Horizontal) – Absolute Severity:** All vectors point to the right. This means points further to the right represent the most extreme events in history (Outliers).
- **PC2 (Vertical) – The Trade-off (Nature of Impact):** This axis reveals a stark contrast:
 - The ***Total Deaths*** vector points UPWARDS (Positive PC2).
 - The ***Total Damage*** vector points DOWNWARDS (Negative PC2).

Analysis of Outliers (The “Monsters” of History)

The Biplot allows us to profile specific historical outliers based on their quadrant position:

- **The “Killer” Outlier (Typhoon Iris, 1964):** Located in the top-right quadrant (High PC1, High PC2). This event is pulled strongly by the *Total Deaths* vector. It represents the era of humanitarian fragility, where a single storm could claim 7,000 lives due to poor infrastructure.
- **The “Bankrupting” Outlier (Typhoon Molave, 2020):** Located in the bottom-right quadrant (High PC1, Low PC2). This event is pulled by the *Total Damage* vector. It represents the modern era: resilient to loss of life, but highly vulnerable to massive financial losses due to expensive coastal infrastructure.

1.2.3 Graph 3: Time Series – The Historical Shift

Trend Analysis

The linear trendline exhibits a clear **negative slope**, confirming a systematic shift in disaster nature:

- **Phase 1: The Humanitarian Crisis Era (Pre-2000):** Most data points (Blue dots) cluster in the positive region ($PC2 > 0$). *Insight:* During this period, Vietnam’s economy was agrarian with low asset density. However, early warning systems were rudimentary. Consequently, disasters were defined by **High Casualties**. The outliers here are tragedies like the 1964 Typhoon Iris.
- **Phase 2: The Economic Burden Era (Post-2000):** Data points (Red dots) migrate significantly to the negative region ($PC2 < 0$). *Insight:* This period coincides with Vietnam’s rapid industrialization and urbanization. The “Four-on-the-spot” policy effectively reduced fatalities (lowering PC2). However, the proliferation of resorts, factories, and urban housing along the Central Coast meant that storms now destroy high-value assets. Disasters are now defined by **High Economic Costs**.

1.3 Limitations of the Study

While PCA successfully reveals structural trends, this study acknowledges certain limitations:

1. **Data Incompleteness:** Historical data (pre-1990) often lacks precise economic damage records. Imputing these missing values with zero may underestimate the economic impact of older events [2].
2. **Event Clustering:** Some historical records may aggregate multiple storms into a single annual entry, potentially inflating the “Severity” (PC1) of that specific year.
3. **Wealth Density Factor:** Although we adjusted for inflation, we did not adjust for the increase in regional GDP. A million-dollar loss in 1964 represented a much larger share of the local economy than in 2024. Normalizing damage by GDP would provide a more accurate measure of true economic resilience.

1.4 Summary

In conclusion, PCA has served as a powerful “decoder” for the history of natural disasters in Central Vietnam. It transformed a chaotic 4-dimensional dataset into a clear historical narrative. The analysis provides quantitative evidence that Vietnam has effectively transitioned from a phase of high human vulnerability to a phase of high asset exposure. This suggests that future disaster management policies must evolve from purely life-saving measures to financial resilience strategies to safeguard the nation’s growing economic wealth.

References

Bibliography

- [1] Group 02, “Principal Component Analysis: Data Dimensionality Reduction,” Linear Algebra Project Report, BK University of Technology, Ho Chi Minh City, 2025.
- [2] Centre for Research on the Epidemiology of Disasters (CRED), “Economic Adjustment Protocol,” *EM-DAT Documentation*, Université catholique de Louvain, Brussels, Belgium. [Online]. Available: <https://doc.emdat.be/docs/protocols/economic-adjustment/>. [Accessed: Dec. 03, 2025].
- [3] General Statistics Office of Vietnam, “Statistical Yearbook of Vietnam 2022,” Statistical Publishing House, Hanoi, 2023.
- [4] The World Bank, “Resilient Shores: Vietnam’s Coastal Development Between Opportunity and Disaster Risk,” Washington, DC: World Bank, 2020. doi: 10.1596/978-1-4648-1605-5.
- [5] J. Shlens, “A Tutorial on Principal Component Analysis,” Google Research, 2014. [Online]. Available: <https://arxiv.org/abs/1404.1100>.